MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
HEALTH, SAFETY & ENVIRONMENT (HSE) DEPARTMENT

CORPORATE SAFETY PROCEDURE: HOT WORK, GAS TESTING & CONFINED SPACE ENTRY
DOC NO: **MRPL-SOP-HSE-0204** | REV: 06 | MANDATORY COMPLIANCE

1. PERMIT TO WORK (PTW) MANDATORY PROTOCOL

No hot work (welding, cutting, grinding, uncertified electrical tools) or confined space entry is permissible within 15 meters of hydrocarbon process equipment without a valid Class-A Hot Work Permit (Form HSE-F01) endorsed by the Shift In-charge (Operations) and the Area Safety Officer.

2. ATMOSPHERIC TESTING LIMITS (GAS TESTING CRITERIA)

Gas / Hazard	Permissible Limit (Cold)	Permissible Limit (Hot Work)	Test Frequency
Combustible Gas (LEL)	< 1.0% LEL	0.0% LEL (Strictly Nil)	Continuous during hot work
Oxygen (O■)	19.5% to 23.5%	19.5% to 23.5%	Prior to entry & every 2 hours
Hydrogen Sulfide (H■S)	< 5.0 ppm	< 5.0 ppm (Nil preferred)	Continuous multi-gas detector
Carbon Monoxide (CO)	< 25.0 ppm	< 25.0 ppm	Prior to entry & post-welding
Volatile Organics (VOC)	< 10.0 ppm	< 5.0 ppm	Calibrated PID detector

3. POSITIVE PHYSICAL ISOLATION (SPADING & BLINDING)

• Double Block and Bleed (DB&B;) is acceptable ONLY for short-duration instrument calibration.
• All line entries, hot work, and vessel entries REQUIRE positive isolation via spectacle blinds or slip spades rated for full piping class design pressure (ASME B16.48). Blinding list sign-off is mandatory before gas freeing.
• Tag-out / Lock-out (LOTO): Padlocks with Master Key retention held by Shift Lead.

4. FIRE PREVENTION & SURVEILLANCE

• Dedicated trained Fire Watch personnel equipped with 50 kg DCP fire extinguisher and continuous pressurized fire hose (10 bar).
• Spark containment habitat / fire-resistant certified blankets (FM Global approved) covering all nearby sewers and drain openings within 15 meters.
• Welding ground cable must be connected directly to the work piece, never to plant structural steel or pipe supports.